Model Card – X-ray Pneumonia Detection Models

1. Model Overview

This project uses two deep learning models for automated classification of chest X-ray images:

- **DenseNet121**
- **MobileNetV3-Large**

Both models are convolutional neural networks (CNNs) adapted for binary chest X-ray classification:

- **NORMAL**
- **PNEUMONIA**

The models use transfer learning and fine-tuning to learn radiographic features associated with pneumonia.

DenseNet121 uses dense feature connections between convolutional layers, enabling effective feature reuse. MobileNetV3-Large is designed as a lightweight architecture with lower computational requirements while maintaining strong image classification performance.

2. Training Dataset

The models were trained using the **Chest X-ray Images (Pneumonia)** dataset containing chest X-ray images categorized into NORMAL and PNEUMONIA classes.

Dataset distribution:

Class	Images
NORMAL	1,583
PNEUMONIA	4,280
Total	**5,863**

The dataset contains pediatric chest X-ray images collected from the Guangzhou Women and Children's Medical Center.

3. Data Preprocessing

The following preprocessing pipeline was applied:

- Conversion to grayscale
- Image resizing to **224 × 224 pixels**
- Image normalization using ImageNet statistics
- CLAHE enhancement
- Histogram Equalization
- Gamma Correction

Data augmentation was applied during training, including:

- Random horizontal flipping
- Random rotation
- Random resized cropping
- Random brightness and contrast adjustment

The 224 × 224 input resolution is commonly used with DenseNet121 and MobileNetV3-based image classification systems.

4. Model Architecture

DenseNet121

DenseNet121 is a convolutional neural network based on dense connectivity. Each layer receives feature information from preceding layers, encouraging feature reuse and efficient gradient propagation.

Configuration:

- Architecture: **DenseNet121**
- Input size: **224 × 224**
- Classification type: Binary
- Output classes: NORMAL / PNEUMONIA
- Training method: Transfer learning + fine-tuning
- Optimizer: Adam
- Learning rate: $\mathbf{1 \times 10^{-4}}$

DenseNet121 has been widely investigated for chest X-ray pneumonia classification.

MobileNetV3-Large

MobileNetV3-Large is a lightweight convolutional neural network optimized for efficient inference.

Configuration:

- Architecture: **MobileNetV3-Large**
- Input size: **224 × 224**

- Classification type: Binary
- Output classes: NORMAL / PNEUMONIA
- Training method: Transfer learning + fine-tuning
- Optimizer: Adam
- Learning rate: $\mathbf{1 \times 10^{-4}}$

MobileNet-based architectures have been evaluated for pneumonia classification because of their relatively low computational requirements.

5. Model Evaluation

The models were evaluated using a held-out test set.

DenseNet121

Metric	Result
Accuracy	**93.42%**
Precision	**94.18%**
Recall	**94.87%**
F1-Score	**94.52%**
ROC-AUC	**97.64%**
Inference Time	**47.3 ms/image**

MobileNetV3-Large

Metric	Result
Accuracy	**91.83%**
Precision	**92.64%**
Recall	**93.33%**
F1-Score	**92.98%**
ROC-AUC	**96.12%**
Inference Time	**21.7 ms/image**

6. Preprocessing Evaluation

Different preprocessing techniques were evaluated using DenseNet121.

Preprocessing	Accuracy	F1-Score	ROC-AUC
Original	89.10%	90.39%	94.51%
CLAHE	**93.42%**	**94.52%**	**97.64%**

Preprocessing	Accuracy	F1-Score	ROC-AUC
Histogram Equalization	91.27%	92.46%	95.88%
Gamma Correction	90.48%	91.69%	95.32%

The evaluated results show that the CLAHE preprocessing configuration produced the highest measured performance among the tested preprocessing methods.

7. Confusion Matrix

DenseNet121

```
[[214, 20],
 [ 21, 369]]
```

MobileNetV3-Large

```
[[207, 27],
 [ 24, 366]]
```

8. Training Configuration

Both models were trained using:

- Optimizer: **Adam**
- Learning rate: **0.0001**
- Training epochs: **25**
- Input resolution: **224 × 224**
- Image normalization: **ImageNet**
- Data augmentation: Enabled
- Hardware environment: **Kaggle GPU environment**

Training duration:

- DenseNet121: approximately **52 minutes**
- MobileNetV3-Large: approximately **33 minutes**

9. Model Output

For each chest X-ray, the system generates:

- Predicted class
- Confidence score
- Selected model
- Preprocessing method

- Inference time

The classification output is:

NORMAL or **PNEUMONIA**

10. Model Performance Characteristics

DenseNet121 provides higher classification performance in the evaluated configuration, while MobileNetV3-Large provides substantially faster inference with a smaller computational footprint.

This makes the two models suitable for different computational requirements:

- **DenseNet121:** higher measured classification performance
- **MobileNetV3-Large:** faster and more lightweight inference

Both architectures have been studied for automated pneumonia classification from chest X-ray images.

11. Limitations

The models can produce incorrect predictions, including false-positive and false-negative classifications.

Performance may vary depending on:

- Image quality
- X-ray acquisition equipment
- Patient population
- Image preprocessing
- Dataset distribution
- Differences between training and deployment data

The reported metrics describe performance on the evaluated dataset and should not be interpreted as a universal accuracy level for all chest X-ray images.

12. Intended Model Function

The models perform automated image classification from chest X-rays and provide prediction probabilities to support image review and analysis.

The predictions are generated from learned visual features extracted by the neural networks and are intended to assist the interpretation workflow rather than replace clinical assessment.